

0l0: A Lattice-Concise Post-Quantum Non-Interactive Zero-Knowledge Proof for a Fixed Relation

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0l0 v0.3.0 prototype

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Abstract

We describe a non-interactive zero-knowledge proof of knowledge for a fixed relation R^* over the cyclotomic ring $R_q = \mathbb{Z}_q[X]/(X^{64} + 1)$, implemented from scratch in safe, dependency-free Rust. Proofs are 4577 bytes, proving takes $\approx 190 \mu s$ and verification $\approx 80 \mu s$ (p50, x86-64). The construction is a two-round Lyubashevsky identification scheme compiled with Fiat-Shamir over a BLAKE3 transcript. This paper states the protocol, the exact parameters, three theorems (completeness, knowledge soundness via extraction, quantified zero knowledge) with their measured evidence, and the explicitly open items. It is a prototype specification, not a security audit.

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1 Relation and parameters

Definition 1.1 (Relation R^*). *Public inputs are an Ajtai matrix $A \in R_q^{4 \times 8}$, a commitment $y = A \cdot w$ and a scalar t . A witness is $w \in R_q^8$ with ternary coefficients such that (i) $A \cdot w = y$, (ii) $\|w\|_\infty \leq 1$, $\|w\|_2 \leq 64$, and (iii) $\langle w, w \rangle = t \pmod{q}$ over centered representatives.*

Symbol	Value	Meaning
D	64	ring degree, $R_q = \mathbb{Z}_q[X]/(X^D + 1)$
q	$1073750017 = 2^{30} + 2^{13} + 1$	prime, $q \equiv 1 \pmod{128}$
$n \times m$	4×8	Ajtai matrix shape over R_q
β, B	1, 64	witness infinity / Euclidean bounds
S, S'	65536, 65472	mask half-width / response bound
w_c	16	challenge weight (nonzero ± 1 coefficients)
τ	2	independent rounds
retries	8	rejection attempts per round
proof	4577 bytes	$1 + 2 \cdot (1024 + 1088) + 288 + 64$

Table 1: Frozen v0.3.0 parameters.

The modulus admits primitive 64-th roots of unity, so all ring products use a from-scratch number-theoretic transform ($W_{64} = 817737403$, verified of exact order 64). Responses satisfy $\|z\|_\infty \leq 65472 < 2^{16}$ and are stored at 17 bits each: $512 \times 17 = 8704$ bits = 1088 bytes exactly, with no padding bits. A single range check on $[64, 131008]$ enforces canonicity and the response bound together.

2 Protocol

Per round $j \in \{0, 1\}$ with caller salt s :

1. Sample a uniform mask $r \in [-S, S]^{512}$ from the XOF stream keyed by the salt, the round index, and the attempt counter.
2. Commit $t_1 = A \cdot r$ and $u = \langle r, r \rangle \pmod{q}$.
3. Derive the challenge c (sparse ternary, exactly 16 nonzero ± 1 coefficients) from the transcript over $(params, A, y, t, s, j, t_1, u)$.
4. Respond $z = r + c \star w$; accept iff $\|z\|_\infty \leq S'$, else retry (at most 8 attempts, else `SamplingExhausted`).

The proof is `version` $\|t_{1,0}\|z_0\|t_{1,1}\|z_1\|u_0 0 u_1 0\|s$ (4577 bytes, version 2). The verifier re-derives both challenges and checks, per round, $A \cdot z = t_1 + c \star y$, the response bound, decoding canonicity, and reserved-zero regions.

The challenge space per round is $\binom{64}{16} \cdot 2^{16} = 2^{64.80}$, giving forking error $\approx 2^{-129.6}$ over two rounds.

3 Theorems

Theorem 3.1 (Completeness). *An honest prover outputs an accepting proof with probability at least $1 - 1.3 \times 10^{-3}$ per proof (else `SamplingExhausted`, and the caller retries with fresh salt).*

Sketch. $A \cdot z = A \cdot r + c \star A \cdot w = t_1 + c \star y$ by R_q -linearity. Since $\|c \star w\|_\infty \leq 16$, per-coefficient acceptance is $(2S' + 1)/(2S + 1)$ and independence gives per-round accept ≈ 0.606 (measured 14-34/40); two rounds with 8 retries fail with probability $\approx 1.3 \times 10^{-3}$ (measured $\leq 2/60$). $\square$

Theorem 3.2 (Knowledge soundness). *From two accepting transcripts sharing (t_1, u) with distinct challenges, either the extractor recovers w^* satisfying all three claims of R^* exactly, or the challenge difference vanishes (discarded; usability $\approx 1 - 64/q$ per pair, measured 0/200 non-invertible). Forgery without extraction needs transcript grinding at $\approx 2^{-130}$ (ROM), each attempt additionally facing a q^{-256} direct-equation barrier.*

Sketch. Same mask r at the fork gives $\Delta z = \Delta c \star w$; invertible Δc yields $w^* = \Delta c^{-1} \star \Delta z = w$ exactly, and all claims are checked on w^* directly — no slack bound is needed. No verifier equation can pin $\langle w, w \rangle = t$ on its own (kernel-ambiguity argument); packing closes through extraction, exhibited at 488/488 on a tiny model with all three claims checked. $\square$

Theorem 3.3 (Zero knowledge, quantified). *Accepted responses are exactly uniform on $[-S', S']^{512}$. For fixed shift $|s| \leq S - S'$, acceptance keeps precisely the values mapping bijectively onto the response range, so the conditional distance is zero in total variation (exhibited by a χ^2 test). Mask commitments are functions of uniform randomness only. Computational hiding rests on mask recovery, estimated at Core-SVP $2^{38} - 2^{42}$ (marginal — not claimed as production-grade).*

Sketch. The uniformity claim is per-coefficient exact arithmetic, lifted jointly by independence. A former packing scalar $v = \langle r, c \star w \rangle$ was removed from the wire after variance analysis showed it encodes $\|cw\|_2$ ($w = 0$ forces $v \equiv 0$); its slots are canonically zero and enforced. $\square$

4 Measured evidence (release day, x86-64)

75/75 tests green (dev and --release); 11/11 official BLAKE3 KATs; an independent Python model produces byte-identical proofs (2/2); a tiny-model extractor recovers witnesses on all 488 usable challenge pairs with 0/2400 direct grinds; exhaustive single-bitflip sweep 36,616/36,616 reject; 20,000-trial random-wire grinding game at 0 accepts; ‘cargo-fuzz’ 14.9M runs clean; full Miri lib run 35/35 green; prover p50 184 μ s, verifier p50 80 μ s.

5 Open items (exact statements)

1. Tail bound on $\|\Delta c^{-1}\|_1$: closed by analysis (the extractor verifies directly, so no tail bound is needed).
2. Closed-form forking-reduction constants: structure proven, constants explicitly out of prototype scope.
3. Hiding beyond conditional uniformity: superseded (responses are exactly uniform); computational leg marginal by measurement.
4. Gaussian sampling, BKZ-simulation depth, external audit, lab side channels, NIST parameter blessing: roadmap or out-of-scope items, tracked in ROADMAP.md.

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